

DSSM IMPORTANT TOPICS

Unit -1

PCA Problems

Finding Covariance Matrix

SVD

Gram Schmidt process

Unit 2

PMF and PDF Problems

Binomial Distribution

Normal Distribution Problems

Uniform Distribution

Unit 3

TPM Problems [Long question Problems]

Show that the Markov chain is irreducible, Regular, ergodic

Long run Probability

Unit 4

Conditional Distribution problems

Marginal Distribution for two variables problems

Unit 5

Least square Method

Find the maximum or minimum

Maximum and Minimum for two variables

Graph Problems

Method of Lagrangian problems (Let me post Question)

Unit 5

Lagrangian function

The association between the slope of the function and slopes of the constraints relatively leads to a reformulation of the initial problem and is called the **Lagrangian function**. Thus, the Lagrange method can be summarized as follows:

To determine the minimum or maximum value of a function $f(x)$ subject to the equality constraint $g(x) = 0$ will form the Lagrangian function as:

$$\mathcal{L}(x, \lambda) = f(x) - \lambda g(x)$$

Here,

$\mathcal{L}$ = Lagrange function of the variable x

λ = Lagrange multiplier

Lagrange's Multipliers Method

There are multiple ways to define the method of Lagrange multipliers. Let's learn one by one.

Suppose f and g are two functions such that they both have continuous partial derivatives.

Let $(x_0, y_0, z_0) \in S := \{(x, y, z) : g(x, y, z) = 0\}$ and $\nabla g(x_0, y_0, z_0) \neq 0$.

If the function f contains a local minimum or local maximum at a point (x_0, y_0, z_0) then there exists $\lambda \in \mathbb{R}$ such that:

$$\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0)$$

To determine the extremum points, we generally consider the below equations:

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z)$$

$$g(x, y, z) = 0$$

Question:

Find the maximum and minimum values of the function $f(x, y) = xy$ subject to the constraint $g(x, y) = 4x^2 + y^2 = 8$.

Solution:

Given,

$$f(x, y) = xy$$

$$g(x, y) = 4x^2 + y^2 = 8$$

As we know, to find the maximum and minimum values for the function $f(x, y)$ subject to the constraint $g(x, y) = 0$ we solve the following equations.

$$(\partial f / \partial x) + \lambda (\partial g / \partial x) = 0$$

$$(\partial f / \partial y) + \lambda (\partial g / \partial y) = 0$$

$$g(x, y) = 0$$

Let's estimate the partial derivative of the first order.

$$\partial f / \partial x = (\partial / \partial x) (xy) = y$$

$$\partial f / \partial y = (\partial / \partial y) (xy) = x$$

$$\partial g / \partial x = (\partial / \partial x) (4x^2 + y^2) = 8x$$

$$\partial g / \partial y = (\partial / \partial y) (4x^2 + y^2) = 2y$$

Substituting all these values in the above set of equations, we get;

$$y + \lambda(8x) = 0$$

$$y = -\lambda(8x) \dots\dots(1)$$

$$x + \lambda(2y) = 0$$

$$x = -\lambda(2y) \dots\dots(2)$$

$$4x^2 + y^2 = 8 \dots\dots(3)$$

Let λ is not equal to 0.

Dividing equation (1) by (2), we get;

$$y/x = -\lambda(8x)/-\lambda(2y)$$

$$y/x = 4x/y$$

$$\Rightarrow y^2 = 4x^2 \text{ \{by cross multiplication\}}$$

$$\Rightarrow y = \pm 2x$$

Substituting $y = \pm 2x$ in equation (3), we get;

$$4x^2 + (\pm 2x)^2 = 8$$

$$4x^2 + 4x^2 = 8$$

$$8x^2 = 8$$

$$\Rightarrow x^2 = 1$$

$$\Rightarrow x = \pm 1$$

Let us write the values of the point (x, y) by taking $x = \pm 1$ and $y = \pm 2x$

	$y = 2x$	$y = -2x$
$x = 1$	$2; (x, y) = (1, 2)$	$-2; (x, y) = (1, -2)$
$x = -1$	$-2; (x, y) = (-1, -2)$	$2; (x, y) = (-1, 2)$

Now, we need to calculate the values of the given function $f(x, y) = xy$ at all these points.

$$\text{So, } f(1, 2) = (1)(2) = 2$$

$$f(1, -2) = (1)(-2) = -2$$

$$f(-1, 2) = (-1)(2) = -2$$

$$f(-1, -2) = (-1)(-2) = 2$$

Therefore, we can get maximum and minimum values at two points each subject to the given constraint. That means,

At $(1, 2)$ and $(-1, -2)$, we get the maximum value for f is 2.

At $(1, -2)$ and $(-1, 2)$, we get the minimum value for f is -2.

Practice Problems

1. Identify the maximum and minimum values of the function $f(x, y) = x^2 + y^2$ subject to $x + 2y - 1 = 0$.
2. Find the extreme values of the function $f(x, y, z) = 2x + 3y + z$ subject to $x^2 + 2y^2 + 3z^2 = 1$.
3. Find the maximum of $f(x, y) = 5x - 3y$ subject to $g(x, y) = x^2 + y^2 = 136$.